5 Feminist Perspectives

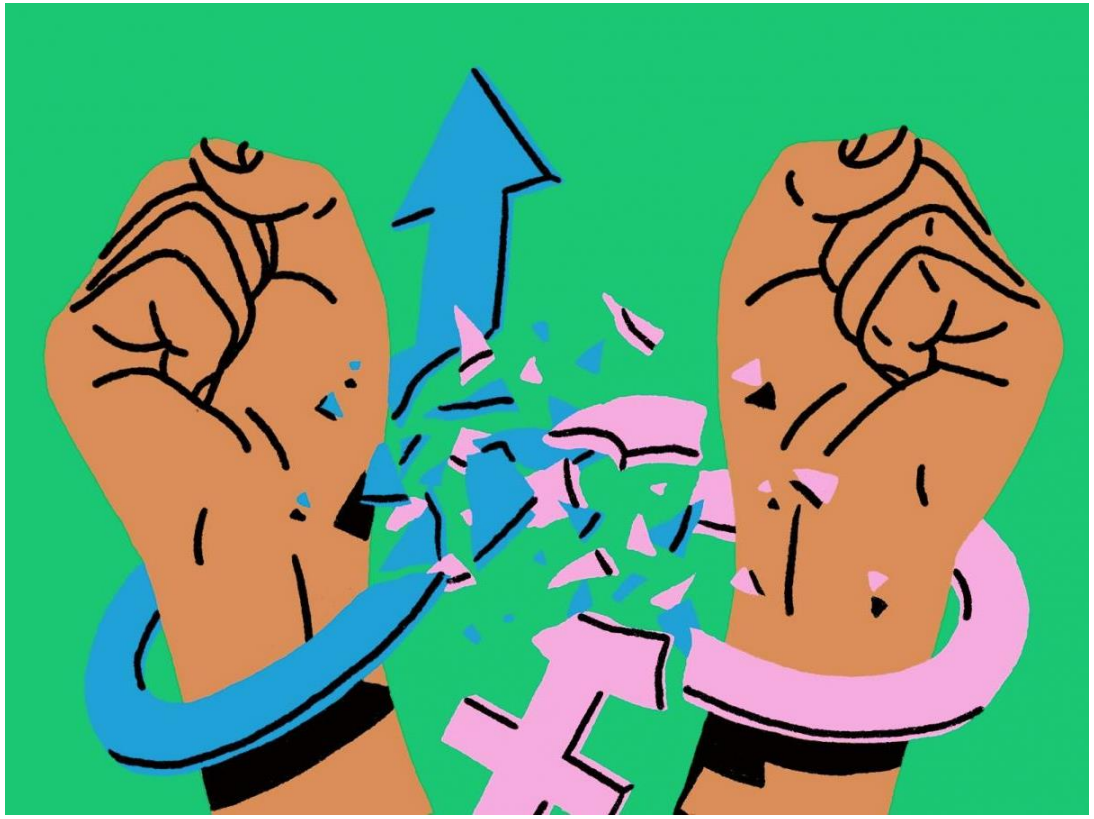
GEB2508 AI and Society

Review of Last Class

- Foucauldian Perspectives on (Scientific) Knowledge
  - Power and Knowledge
  - Truth and Discourse
  - Biopower and Governmentality
- Foucauldian Perspectives on Technology
  - The Panopticon
  - Surveillance Technology

Content

- Feminism: Theory & Reality
  - Sex and Gender
  - Determinism vs. Constructivism
  - Gender Inequality
- Feminist Perspectives on Science and Technology
  - Gender Representation and Normativity in STEM
  - Gendered Scientific Knowledge
  - Technology Through Feminist Perspectives



Basic Concepts

Sex

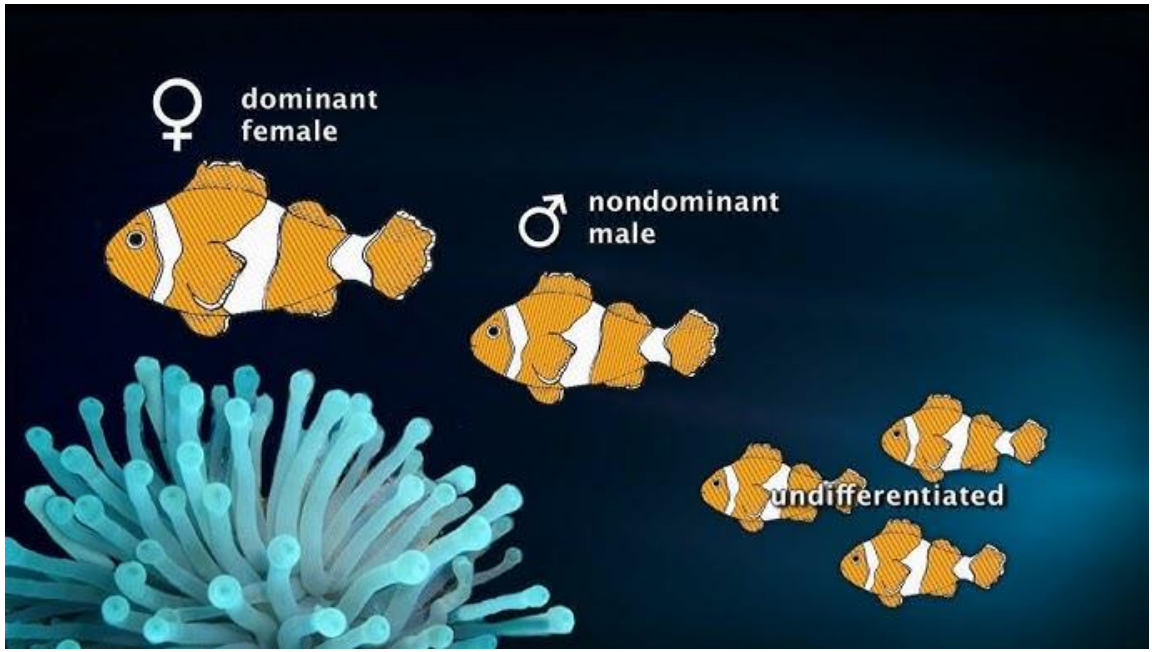
- A **biological** distinction based on roles in biological reproduction
- Including characteristics like chromosomes, anatomy, and hormone levels
- Sex typing: male and female
- Non-binary and not fixed

Gender

- A **socially** constructed distinction between roles and expectations associated with sex
- Gender typing: masculine and feminine
- Non-binary; exists on a spectrum
- Fluid and influenced by socio-cultural norms

Non-Binary Sex in Animal Species

Clownfish



White-throated sparrow

Non-Binary Human Sex

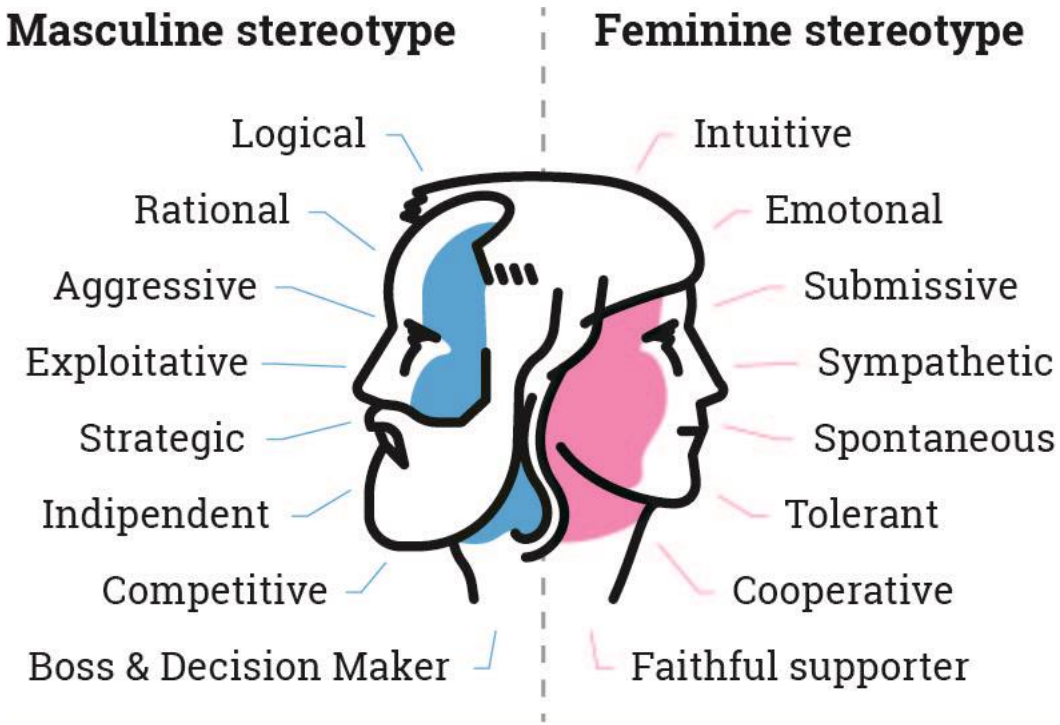
- Chromosomal variations
  - In addition to XY and XX: XO, XXY, XYY, etc.
- Anatomical variations
  - Intersex individuals with ambiguous genitalia
- Hormonal variations
- These variations are not anomalies but **natural** expressions of human **diversity**.

Biological Determinism

- Nature vs. nurture
- Gender roles are biologically determined and are therefore fixed and unchangeable.
- Throughout history, studies on sex differences have been used to provide **scientific justification** for gender inequality.
- Research on the "gay gene" (2019)

Social Construction of Gender

- Gender as a **process** and part of social structure.
- Masculine and feminine stereotypes oppress **both** women and men by enforcing rigid expectations of conformity.
- Historical and cultural variability: expectations of gender roles **vary** across societies and **evolve** over time.



Gender Signifiers

- Clothing: Dresses and skirts are conventionally feminized, while pants are masculinized
- Hairstyles: Long hair is often read as feminine, whereas short hair as masculine.
- Color coding: "Pink for girls, blue for boys."
  - Pink tax

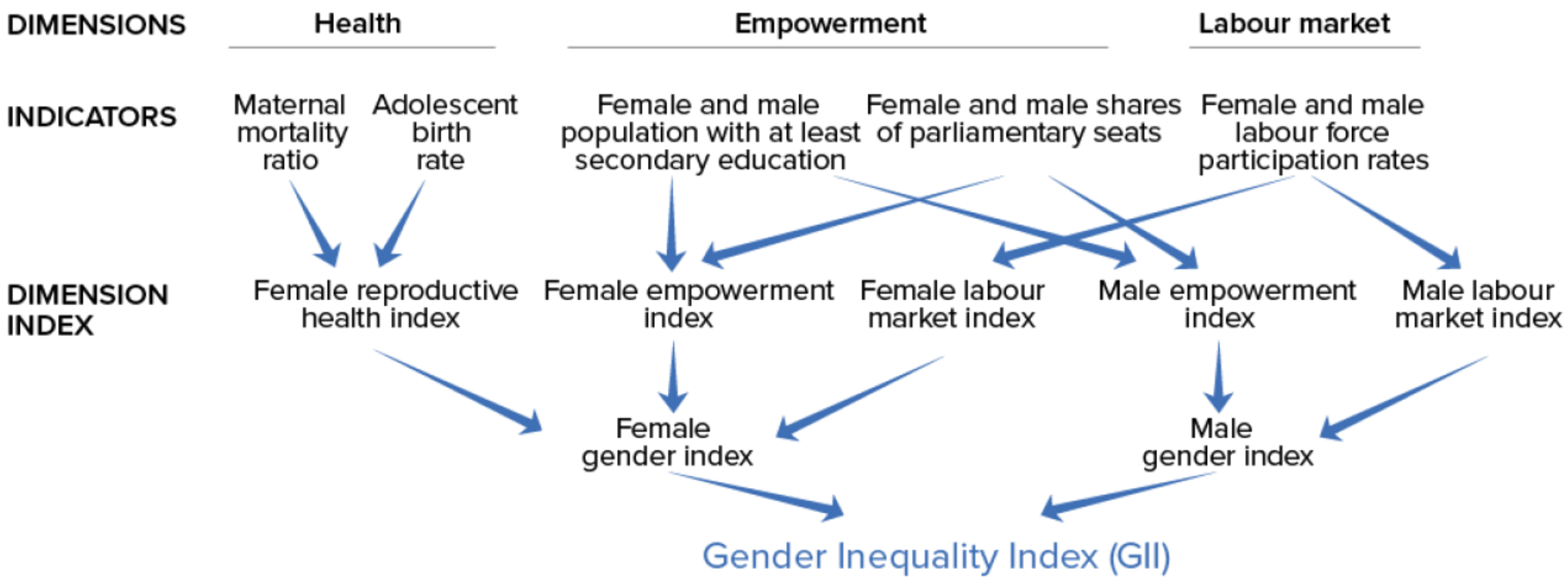
The Second Sex

- *The Second Sex* (1949) is written by the French philosopher Simone de Beauvoir.
  - A seminal work in feminist philosophy and gender studies
- Man is considered the default, while woman is considered the "**other**."
  - "One is not born, but rather **becomes**, a woman."
  - "Man is defined as a human being and woman as a female."

Happy International Women's Day to the cute daughters, sweet sisters, lovely girlfriends, darling wives and adorable mothers!

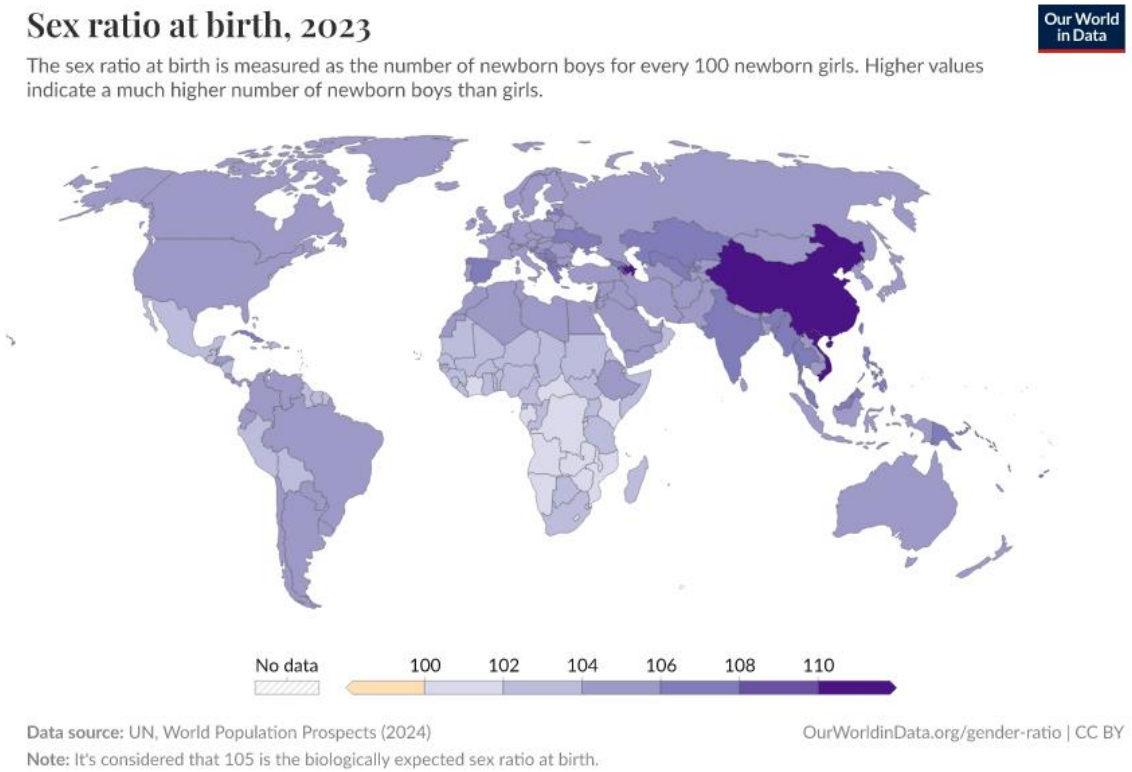
Best Regards,
HSS
The School of Humanities and Social Science

Gender Inequality



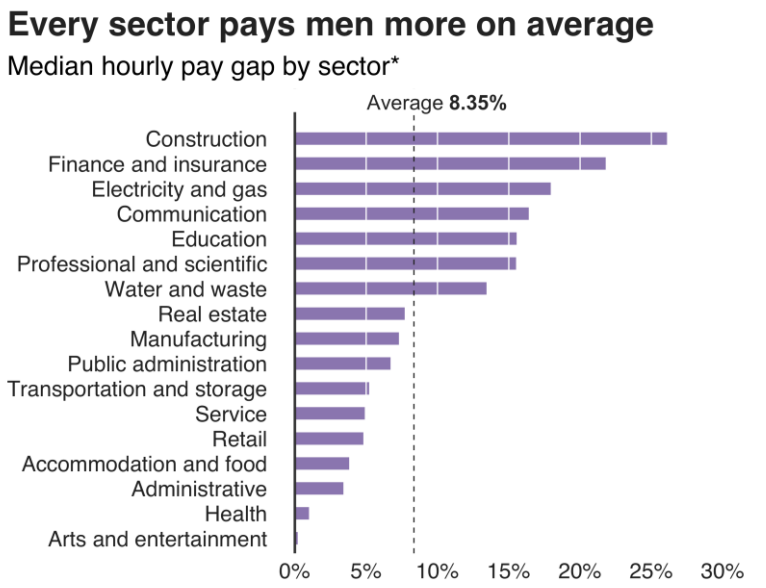
The Gender Inequality Index (GII) is a measure of gender inequality developed by the United Nations Development Programme (UNDP).

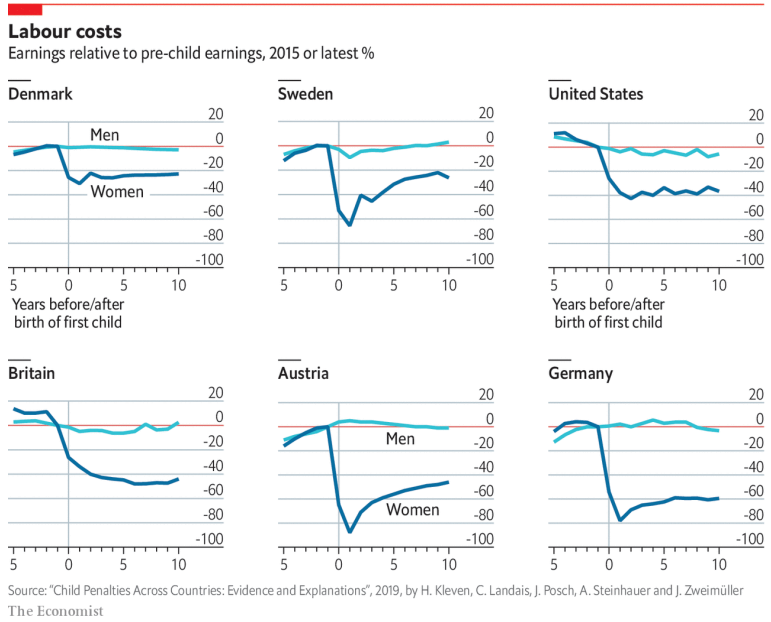
Right to Life



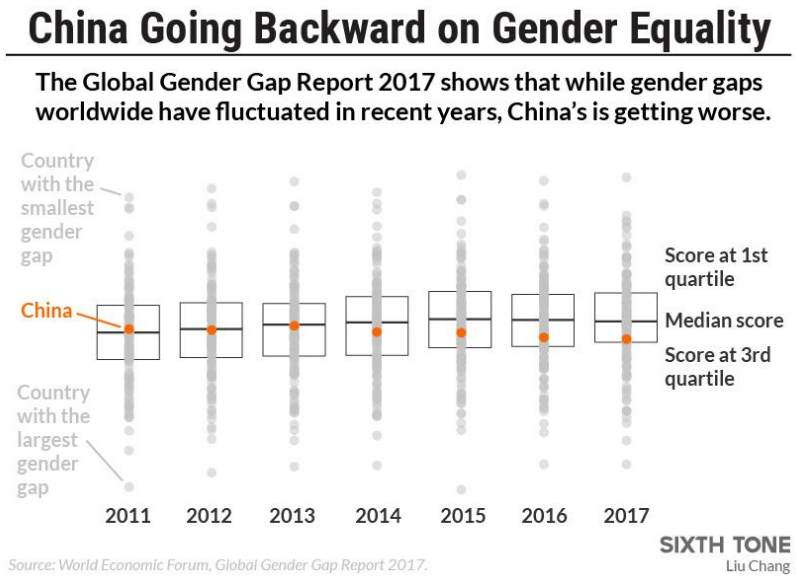
Gender Pay Gap

- Definition: difference between women's and men's earnings in the workforce
- **Unpaid** domestic work
  - Women do more than three-quarters of unpaid domestic work worldwide.
- Birth penalty
  - Women with children are discriminated against in the job market; men are not.
  - Motherhood can affect job choices.





Does Economic Growth Guarantee Gender Equality?



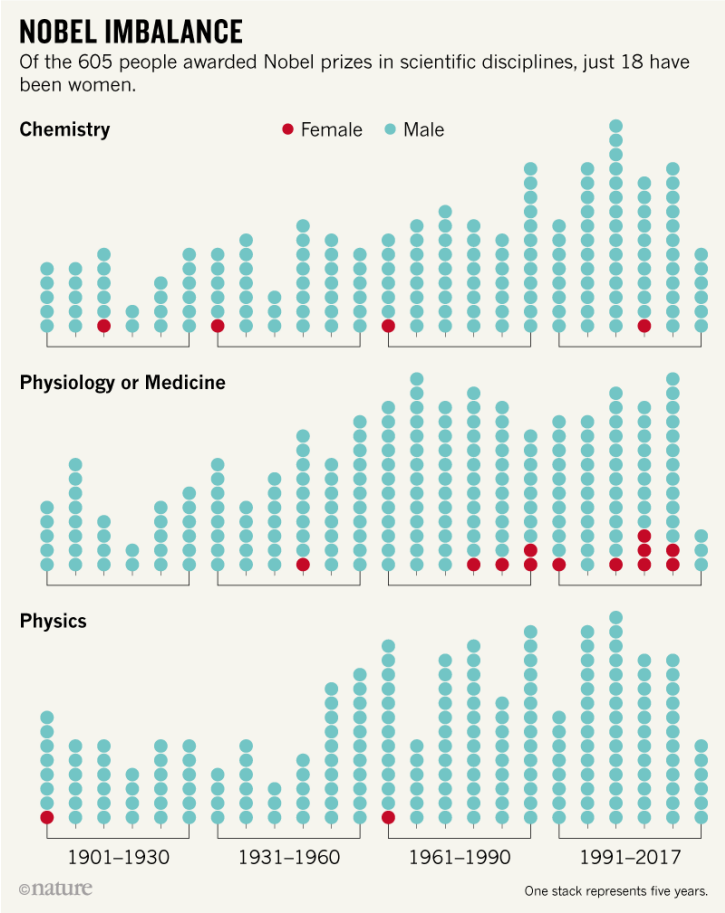
Gender equality ranking

This year	(Last year)	146 countries	Gender gap score
1	1	Iceland	Improvement ↗
2	3	Norway	↗
3	2	Finland	↗
4	4	New Zealand	↗
5	5	Sweden	Deterioration ↘
6	10	Germany	↗
7	7	Nicaragua	↗
8	8	Namibia	↘
9	11	Lithuania	↗
10	14	Belgium	↗
43	27	United States	↘
105	99	South Korea	↘
107	102	China	↘
125	116	Japan	↘
146 (lowest)	146	Afghanistan	↘

Source: Global Gender Gap Report 2023 by World Economic Forum

Under-Representation of Women in STEM

- The STEM gap: men substantially outnumber women in most STEM fields.
- Explicit and implicit biases within the workplace and persistent challenges surrounding life-work balance have driven women out of STEM fields.
  - Gender stereotypes
  - Male-dominated cultures
  - Fewer role models



Structural Obstacles: The Case of Marie Curie

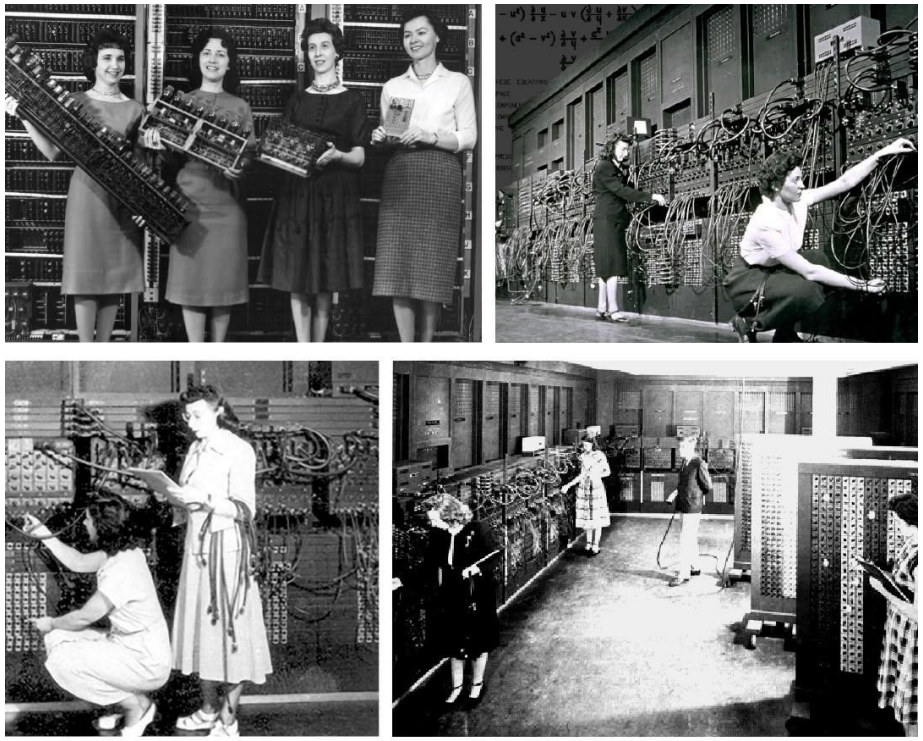
- Restricted access to education
  - Universities in Poland did not accept female students or researchers.
- Gender bias and institutional discrimination in academia
  - Initial nomination of the 1903 Nobel Prize in Physics excluded her name.
  - She was denied membership to the French Academy of Sciences in 1911.
- Work–family pressures

Under-Recognition of Women's Contributions

- Rosalind Franklin and the discovery of DNA
- Lise Meitner and nuclear fission
- Jocelyn Bell Burnell and the discovery of Pulsars
- Rosalind Lee and the discovery of microRNAs
- ……

Women and the History of AI Development

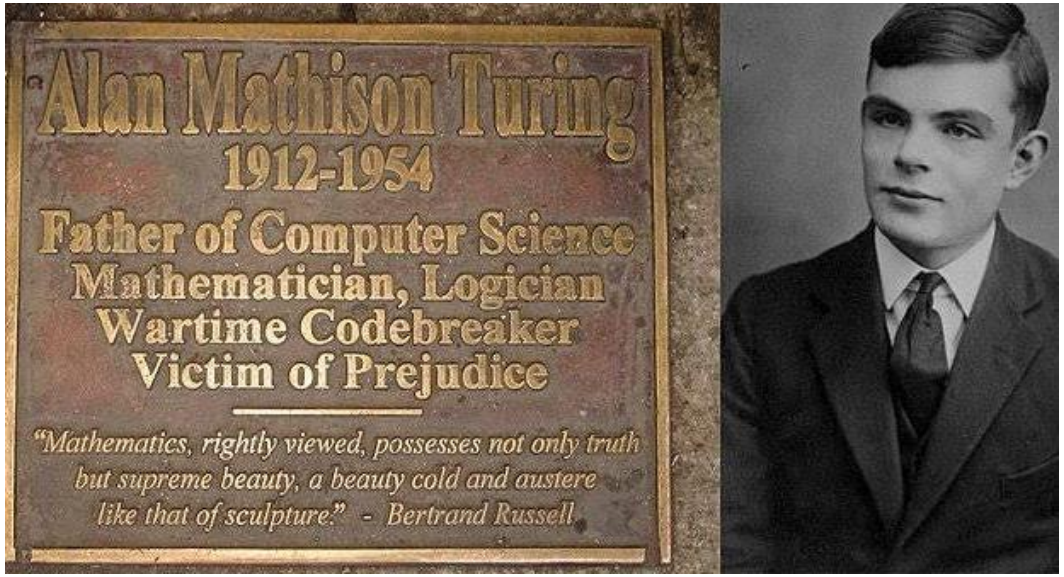
- Women have played foundational but **under-recognized** roles in computing and AI - as theorists, programmers, and laborers.
  - Ada Lovelace (1815-1852): an early pioneer of programming
  - ENIAC programmers (1940s)
- AI developed through **gendered** divisions of labor



THE PROGRAMMING PIONEERS OF ENIAC

Turing through Feminist and Queer Lenses

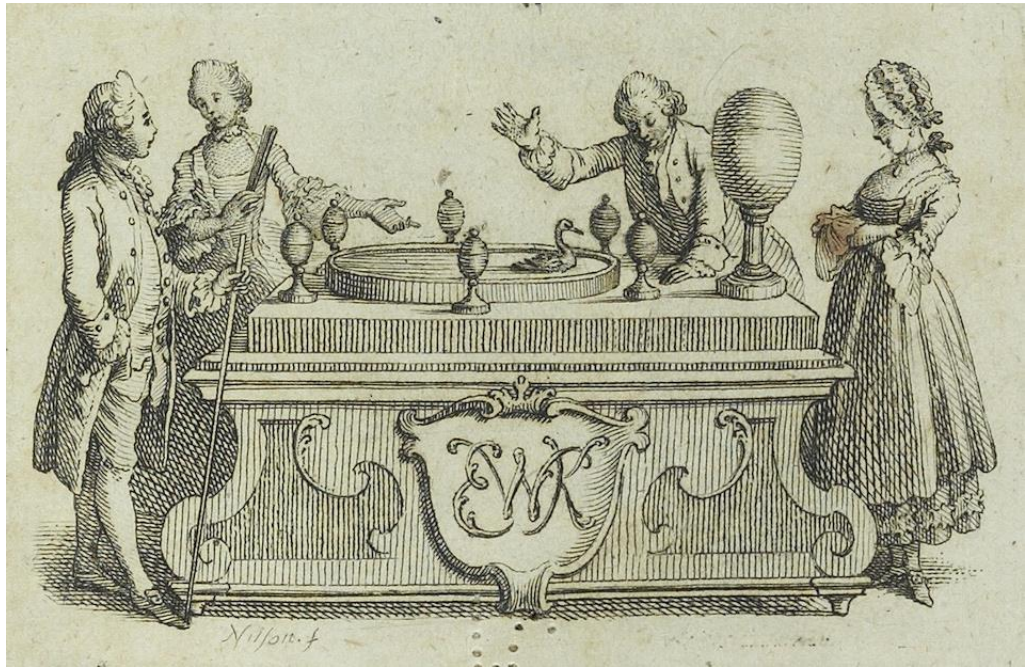
- Alan Turing: pioneer of computer science and AI; helped defeat Nazi Germany
- Persecution for homosexuality: personal tragedy or **structural** problem?
  - Homosexuality as **illness**
- The ideal of **scientific masculinity**: emotionally restrained, rational, heterosexual



Turing convicted for indecency, 1952
Turing found dead of cyanide poisoning, 1954
Turing received a posthumous royal pardon, 2014

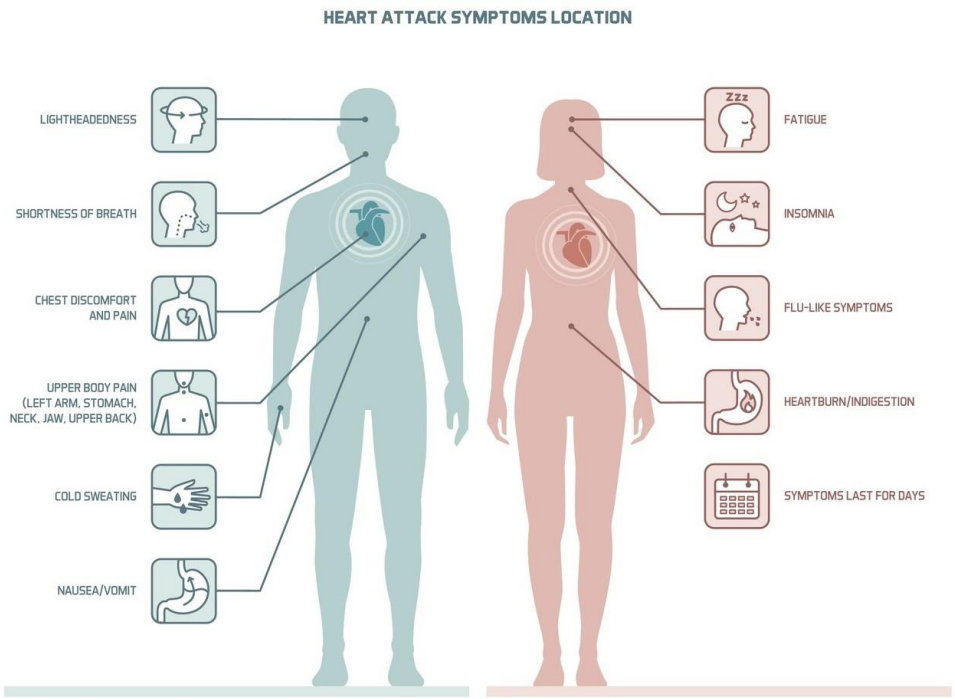
Gendered Scientific Knowledge

- Gender biases and assumptions in scientific research: production of knowledge that **reflects** and **reinforces** gender inequality.
- Since the rise of modern science, the field has been **historically** male-dominated, which influences the production, dissemination, and application of scientific knowledge.



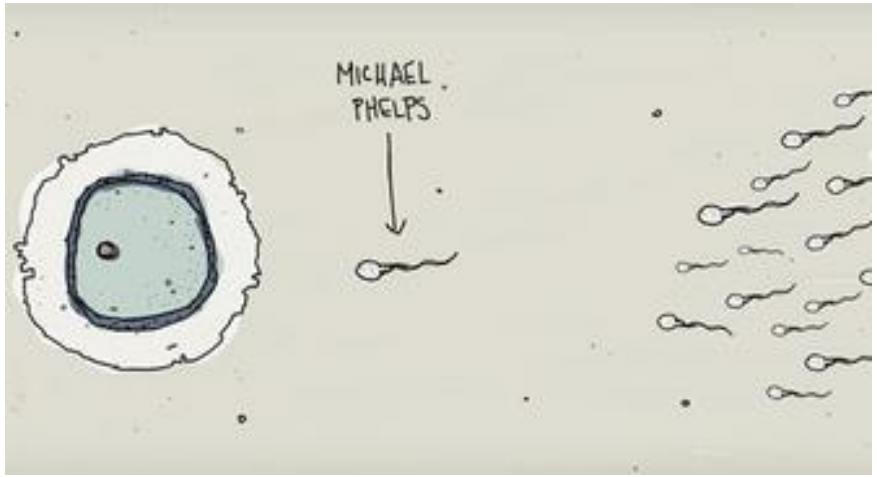
The Case of Medical Research

- Male-as-default bias: clinical trials and medical research have historically centered male bodies as the **normative** standard.
  - Higher rates of misdiagnosis and underdiagnosis among women
  - All-male cohorts of lab mice
- Gendered assumptions in science
  - Pathologization of women's emotions



The Fertilization Story

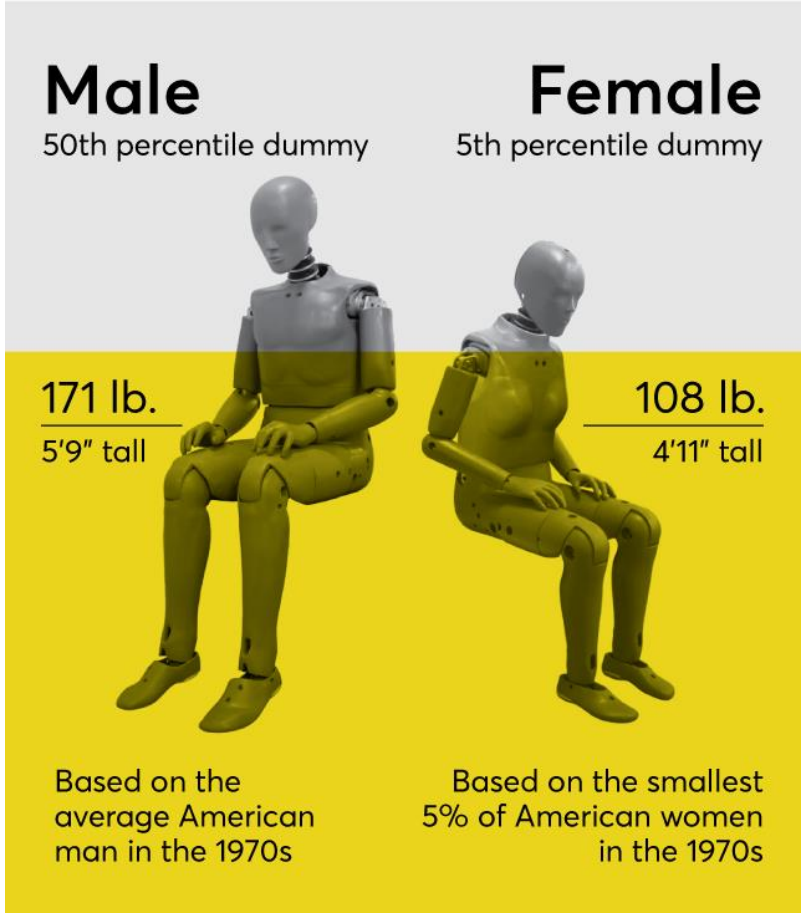
- How do science textbooks use gendered language when describing the respective roles of sperm and egg in fertilization?
- What gender-neutral alternatives could be used to explain fertilization without importing stereotypical roles?
- Who might benefit from perpetuating the traditional narrative of fertilization?



Inventions That Have Empowered Women

Gendered Technological Artifacts

- The 50th-percentile (white) male as the **default** subject of design
- Automotive safety: cars are frequently optimized for a male “reference man,” contributing to women’s substantially higher risk of serious injury
- Air-conditioned workplaces are often calibrated to male metabolic norms
- Speech-recognition bias



Household Technology

- Labor-saving household technologies
  - Vacuum cleaner (1913), washing machine (1916), refrigerator (1918), microwave oven (1973)
- **Reorganize** rather than eliminate domestic labor, raising household standards and expanding **expectations** of motherhood
- Has household technology emancipated women?

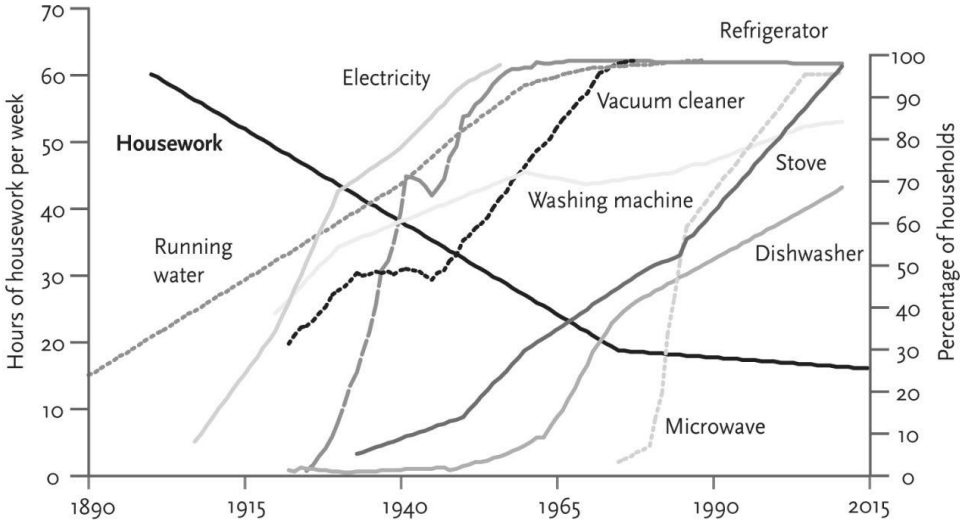
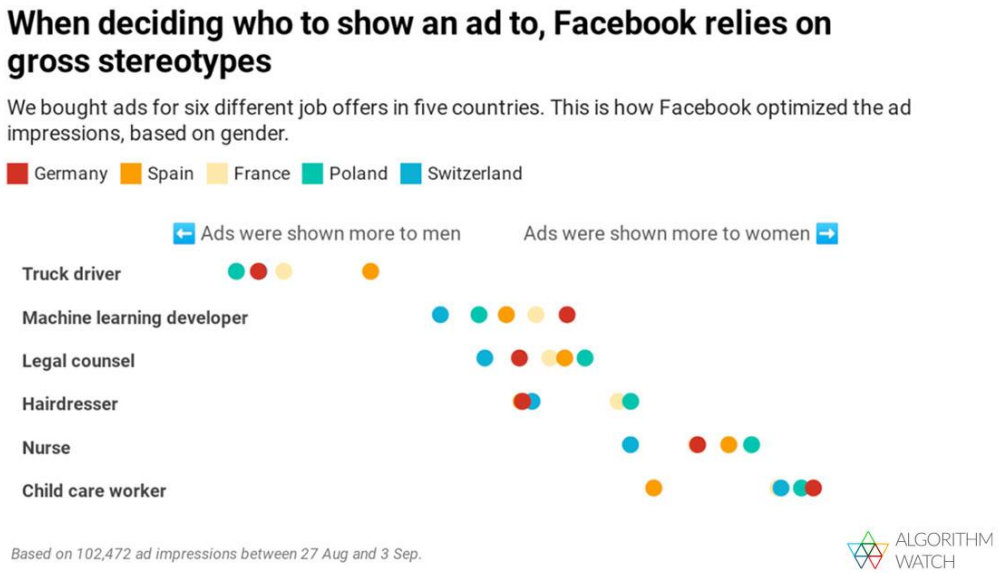
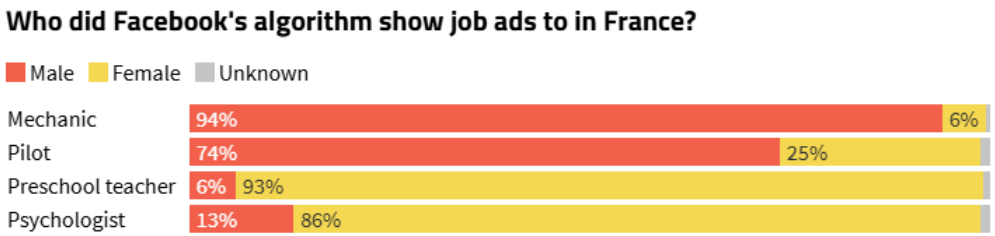


Figure 17-3: Utilities, appliances, and housework, US, 1900–2015

Sources: Before 2005: Greenwood, Seshadri, & Yorukoglu 2005. **Appliances, 2005 and 2011:** US Census Bureau, Siebens 2013. **Housework, 2015:** *Our World in Data,* Roser 2016t, based on the American Time Use Survey, Bureau of Labor Statistics 2016b.

The Kitchen Debate (1959)

- Between Nixon and Khrushchev at the American National Exhibition in Moscow
  - Cold War context: technology used as evidence of **ideological** superiority
  - Household appliances framed as symbols of progress and freedom
- Kitchen designed for a heterosexual, nuclear family, while housework remains **feminized** and unpaid

Gender Bias and Discrimination in AI

- Gender bias in AI
  - Facial recognition
  - Hiring and recruitment algorithm
- Gendered design of AI agents
  - Feminized voice assistants: obedience, servility, affective
- Data as a site of gendered power
  - Historical and structural encoding